

Local Limits of Small-World Networks

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Small World Networks

Many real-world networks—social, biological, and technological—exhibit small-world properties:

- **high local clustering** (friend of a friend is a friend!)
- **short average distances** (it's a small world!)

Goal: Characterizing the limiting distribution of the local motifs as networks grow.

Watts-Strogatz (WS) model

The WS model starts from a k -regular ring and rewires each edge independently with probability ϕ to a random node [4], giving $WS(n, k, \phi)$.

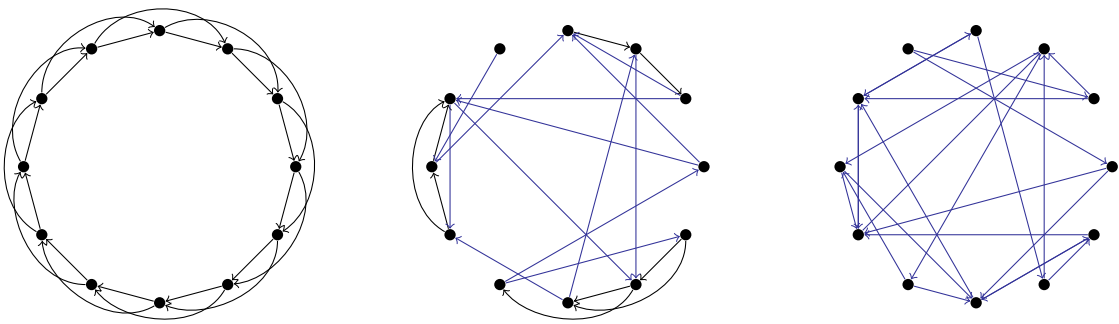


Figure: $WS(12, \phi, 2)$ increasing randomness from left to right: for $\phi = 0$, $0 < \phi < 1$, $\phi = 1$ respectively. Colored edges correspond to shortcuts while black edges correspond to ring edges. Arrows indicate the directions of edges in the Watts-Strogatz process but the final graphs are undirected.

Kleinberg model

The Kleinberg model embeds nodes on an $n \times n$ grid and adds q long-range shortcuts from each node, where the probability of a shortcut decays as a power-law distribution of lattice distance with exponent ℓ [3], giving $K(n, q, k, \ell)$.

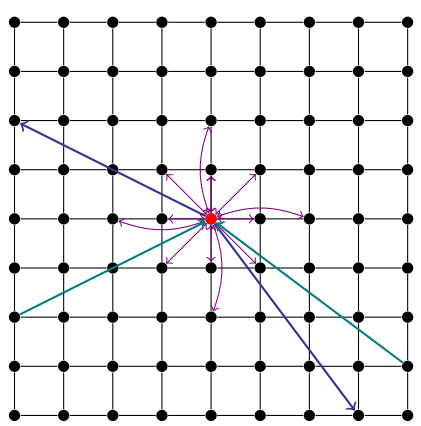


Figure: $K(9, 2, 2, \ell)$ for some ℓ . For simplicity, we plot the directed edges that only belong to the center node and colored edges based on the type.

Theory of Graph Limits

Local convergence (also known as **Benjamini-Schramm convergence**) is a central technique in random graph theory [1]. It describes the notion that a (sparse) graph, seen from the perspective of a node chosen uniformly at random from all nodes, behaves like a certain—often simpler—limiting graph as the number of nodes goes to infinity.

Definition (Local Convergence in Probability)

Let $(G_n, M_n)_{n \geq 1}$ be a sequence of random graphs. Then $(G_n, M_n)_{n \geq 1}$ converges locally in probability to $(G, o, M) \sim \mu$, when for any $\epsilon > 0$, all integers $r > 0$, and any marked rooted graph $H_* = (H_*, o_*, M_*)$,

$$\frac{1}{n} \sum_{o_n \in V(G_n)} 1_r^\epsilon((G_n, o_n) \simeq H_*) \xrightarrow{\mathbb{P}} \mu(1_r^\epsilon((G, o) \simeq H_*))$$

where 1_r^ϵ is defined as the indicator that the r -neighborhood of (G_n, o_n) and (H_*, o_*) are isomorphic, and the distance of the corresponding marks is bounded by ϵ . Formally,

$$1_r^\epsilon((G_n, o_n) \simeq (H_*, o_*, M_*)) = 1 \left(B_r(G_n, o_n) \simeq B_r(H_*, o_*) \text{ and } \exists \pi \text{ such that, } \max_{u \in V(B_r(H_*, o_*))} d_\Xi(M_n(u), M_*(\pi(u))) < \epsilon \right. \\ \left. \max_{uv \in E(B_r(H_*, o_*))} d_\Xi(M_n(u, v), M_*(\pi(u, v))) < \epsilon \right)$$

where π runs over all isomorphisms between $B_r(G_n, o_n)$ and (H_*, o_*) .

Remark. $\arg\max_{r, 1/\epsilon} 1_r^\epsilon(\cdot, \cdot)$ gives a natural metric that makes the set of rooted graphs with bounded degree **separable** and **Polish**, allowing us to define a measure on it and have convergence on this space.

Theorem (Local Limit of the WS model)

For any integer $k \geq 1$, Watts-Strogatz's model converges locally in probability to the Full k -Fuzz exemplified below, i.e., $WS(n, k, \phi) \xrightarrow{\mathbb{P}} WS(k, \phi)$ as $n \rightarrow \infty$.

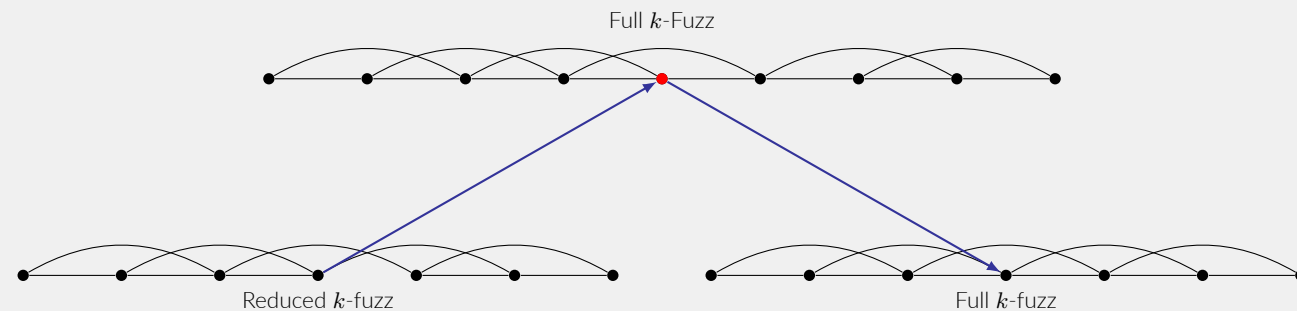


Figure: Full k -fuzz (for $k = 2$). The outgoing shortcut from the root (shown in red) connects to another full k -fuzz, while the incoming shortcut connects to a reduced k -fuzz.

Theorem (Local limit of the Kleinberg model ($\ell \leq 2$))

For any integers k and $q > 0$, $\ell \leq 2$, the Kleinberg model $K(n, q, k, \ell)$ converges locally in probability to $K_{\leq}(q, k, \ell)$.

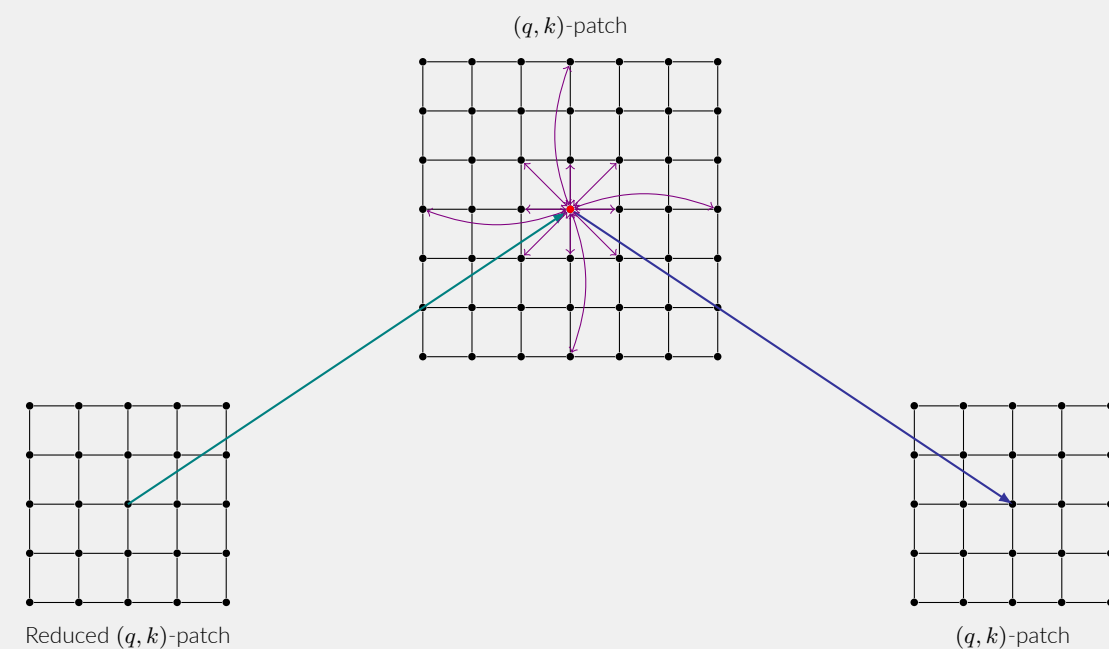


Figure: The local limit structure of $K(n, q, k, \ell)$ for $\ell \leq 2$: (q, k) -patch (for $k = 2$). The outgoing shortcut from the root (shown in red) connects to another (q, k) -patch, while the incoming shortcut connects to a reduced (q, k) -patch. (For simplicity, we plot the directed edges that only belong to the root and colored edges based on the type.)

Remark: Hidden random tree

- When $k = 1$, Watts-Strogatz's model converges locally in probability to a multi-type branching process with 4 types.
- For $\ell \leq 2$, the subgraphs induced by the shortcuts in $K_\ell(q, k, \ell)$ converge in probability to a multitype branching process.

Theorem (Local limit of the Kleinberg model ($\ell > 2$))

For $\ell > 2$, the local limit of the Kleinberg model exhibits a fundamentally different behavior compared to the case $\ell \leq 2$. In this regime, shortcuts are local, connecting nodes within a bounded lattice distance...

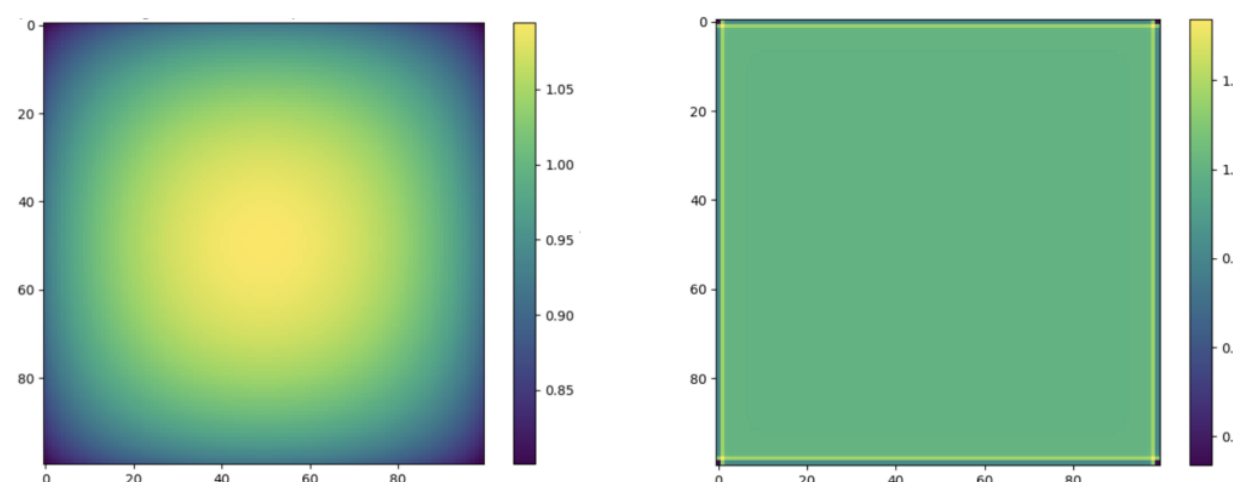


Figure: (left) Expected number of incoming shortcuts for $n = 100$, and $\ell = q = 1$. (right) Expected number of incoming shortcuts for $n = 100$, $q = 1$ and $\ell = 5$.

Proof sketch

1. We analyze rooted neighborhoods via BFS exploration with **marks** encoding edge types for the WS model and normalized node location in the grid for the Kleinberg model, which concentrates from discrete to continuous laws.
2. The **first-moment** argument **couple**s the exploration to the limiting structure, identifying the expected local structure; the **second-moment** bound ensures concentration and convergence in probability.

Algorithmic Applications

By local convergence, many structural and dynamical quantities in small-world networks admit precise limiting values and constant-time estimators.

Meta-Principle: Constant-Time Estimation.

A function f is r -local if $f(G, o)$ depends only on the radius- r neighborhood $B_r(G, o)$ (including marks). If $G_n \xrightarrow{\mathbb{P}} \mu$ and f is bounded and r -local, then

$$f(G_n, o_n) \xrightarrow{\mathbb{P}} f(G, o), \quad \mathbb{E}f(G_n, o_n) \rightarrow \mathbb{E}f(G, o).$$

For $\theta(G) = \mathbb{E}_{G \sim \mu}[f(B_r(G, o))]$, the estimator

$$\hat{\theta}_m = \frac{1}{m} \sum_{i=1}^m f(B_r(G, o_i))$$

with $m = O(\epsilon^{-2} \log(1/\delta))$ satisfies $|\hat{\theta}_m - \theta(G)| \leq \epsilon$ with probability at least $1 - \delta$.

1. Clustering Coefficient. For $v \in V(G)$,

$$C(v) = \frac{\Delta_G(v)}{d_v(d_v - 1)}, \quad C_{\text{global}} = \frac{\sum_v \Delta_G(v)}{\sum_v d_v(d_v - 1)},$$

where $\Delta_G(v) = \sum_{u, w \in \partial B_1(v)} 1_{\{uw \in E(G)\}}$ counts twice the triangles through v . Shortcuts make neighborhoods locally tree-like, so limiting clustering depends on ring edges:

$$C_{\text{global}}^{WS} = \frac{3(k-1)}{2(2k-1) + \phi(2-\phi)}(1-\phi)^3, \quad C_{\ell \leq 2}^K = \frac{\Delta_{L_k}}{A^2 - A + 2A\Lambda_m + \Lambda_m^2},$$

with $A = 2k(k+1) + q$ and Λ_m the mean number of incoming shortcuts.

2. PageRank. The normalized PageRank of $WS(n, \phi, k)$ and $K(n, q, k, \ell)$ converges in distribution under local convergence. Because PageRank depends only on finite-radius random-walk return probabilities, it can be computed from local neighborhoods with exponentially decaying truncation error [2].

3. Spanning Trees and Tree Entropy. Under local convergence,

$$\frac{1}{|V(G_n)|} \log \tau(G_n) \rightarrow h(\mathbf{G}) = \mathbb{E}[\log \deg_{\mathbf{G}}(o)] - \sum_{t \geq 1} \frac{1}{t} \mathbb{E}[p_t(o, o)],$$

where $p_t(o, o)$ is the t -step return probability of a simple random walk. In $WS(\phi, k)$, increasing ϕ lowers $p_t(o, o)$ and raises $h(\mathbf{G})$; in $K(n, q, k, \ell)$, $\ell \leq 2$ induces expansion (higher entropy), whereas $\ell > 2$ yields lattice-like behavior.

4. Epidemic and Cascade Dynamics. For an SIR process with parameters (ρ, β, μ) ,

$$\mathbb{E}[X_t^{\text{SIR}}(G_n, o_n)] \rightarrow \mathbb{E}[X_t^{\text{SIR}}(\mathbf{G}, o)].$$

The infected fraction can be estimated using radius- $R(\beta, \epsilon)$ neighborhoods of $m = O(\epsilon^{-2} \log(1/\delta))$ roots. In $WS(\phi, k)$, increasing ϕ accelerates early spread; in $K(\ell \leq 2)$, branching-like expansion amplifies prevalence.

Phase transition and new perspective on decentralised search.

In Kleinberg's greedy routing (always move to the neighbor closest to the target),

efficiency changes sharply at $\ell = 2$: $T_n = \begin{cases} O((\log n)^2), & \ell = 2, \\ \Theta(n^\alpha), & \ell \neq 2, \end{cases}$ for some $\alpha > 0$. Mirroring

this phase transition, in the marked local limit, each shortcut's mark encodes its scaled length.

- When $\ell > 2$, marks cluster near zero—shortcuts are local and navigation is slow.
- When $\ell < 2$, marks are heavy-tailed—shortcuts overshoot the target.
- At $\ell = 2$, marks are scale-invariant, producing a hierarchy of sublinear links and a navigable metric space that enables efficient decentralized search.

References

- [1] Itai Benjamini and Oded Schramm, *Recurrence of distributional limits of finite planar graphs*, 2001.
- [2] Remco van der Hofstad, *Random graphs and complex networks*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, 2016.
- [3] J. Kleinberg, *The small-world phenomenon: an algorithmic perspective*, Proceedings of the Thirty-Second Annual ACM Symposium on Theory of Computing (New York, NY, USA), STOC '00, Association for Computing Machinery, 2000, p. 163–170.
- [4] Duncan J. Watts and Steven H. Strogatz, *Collective dynamics of 'small-world' networks*, Nature **393** (1998), no. 6684, 440–442.